



REGULARIZED MACHINE LEARNING vs. TRADITIONAL ECONOMETRICS FOR FORECASTING INDIA'S MACROECONOMIC FUTURE

A Rigorous Out-of-Sample Evaluation of ARIMA, Ridge Regression, and Tree-Based Models
Across GDP Growth, CPI Inflation, and Financial Markets (2008–2025)

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01 ABSTRACT / PROJECT OVERVIEW

Traditional econometric models such as ARIMA are designed for parameter estimation and inference, emphasizing statistically interpretable coefficients (β). However, strong in-sample explanatory power does not guarantee robust out-of-sample predictive performance (y) in noisy economic environments.

This project benchmarks traditional econometrics against machine learning models under a strict “firewall” protocol that eliminates look-ahead bias and future data leakage.

We develop a quarterly macroeconomic forecasting pipeline (2007–2025) leveraging IS–LM interactions to predict GDP growth, CPI inflation, and Nifty 50, and (2) a monthly stock forecasting pipeline (2022–2026) evaluating five major NSE stocks: HDFC Bank, Trent, TCS, Larsen & Toubro, and Titan.

The results show that carefully regularized models can outperform classical baselines in low signal-to-noise settings.

02 METHODOLOGY: FEATURE ENGINEERING & THE “FIREWALL”



2.1 IS–LM SPILLOVER ENGINEERING (ZERO LOOK-AHEAD FEATURES)

- All rolling features created with strict .shift(1) to eliminate look-ahead bias.
- For each macro target, 11 lagged features were engineered to capture IS–LM spillovers:
 - Lagged GDP growth, CPI inflation, Nifty returns
 - Rolling means, volatility, momentum
 - Cross-variable interactions & transmission lags
- All features at time t use only data available at $t-1$.



2.2 THE FIREWALL SPLIT (ZERO LEAKAGE)

A strict chronological train/test split was enforced. No random shuffling. No future contamination.

MACRO PIPELINE (Quarterly)

- Train: 2008 Q3 – 2022 Q4 (~55 obs.)
- Test: 2023 Q1 – 2025 Q1 (9 quarters)

STOCK PIPELINE (Monthly)

- Train: Jan 2022 – Dec 2025 (48 months)
- Test: Jan 2026 – May 2026 (5 months)

Simulates real-world production deployment.



2.3 COVID-19 SHOCK TREATMENT (CRUCIAL DESIGN CHOICE)

- India's ~6.60% GDP drop in 2020 is an exogenous, unexplainable shock.
- 2020 data removed from training targets (y) to avoid penalizing models.
- 2020 data retained in feature matrix (X) so lagged inputs can capture rebound dynamics in 2021.

03 MODEL SUITE & EXTREME REGULARIZATION



3.1 ARIMA BASELINE (ECONOMETRIC MODEL)

- Tuned via automated AIC grid search.
- Differencing (1) used for non-stationary series.
- Provides robust stochastic baseline.



3.2 RIDGE REGRESSION (L2 REGULARIZATION)

- Preferred over LASSO due to high collinearity.
- Extreme alpha grid: 1, 10, 100, 500, 1,000, 5,000, 10,000 to force coefficient shrinkage.
- Controls variance and improves generalization.



3.3 TREE-BASED MODELS (SHALLOW TREES)

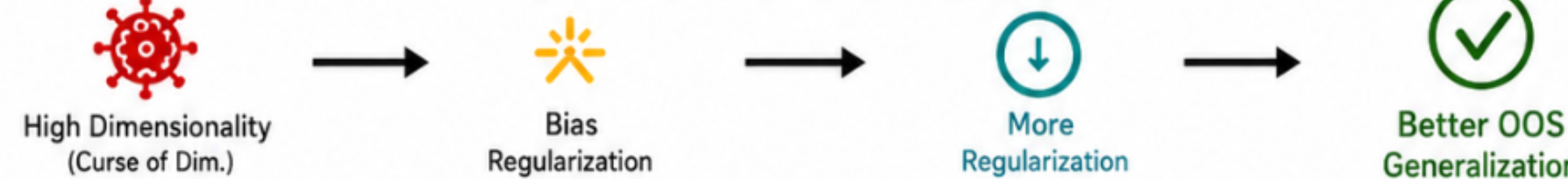
- Random Forest, Decision Tree, Gradient Boosting
- Max depth = 3 across all models.
- Prevents “fitting on conundrum” of memorizing idiosyncratic noise in ~55 observations.
- Reduces variance and overfitting risk.



3.4 ENSEMBLE (VOTING REGRESSOR)

- Averages predictions of Ridge, RF, DT, GB.
- Lowers variance by combining diverse model structures.
- Defensive strategy—rarely best, but most stable.

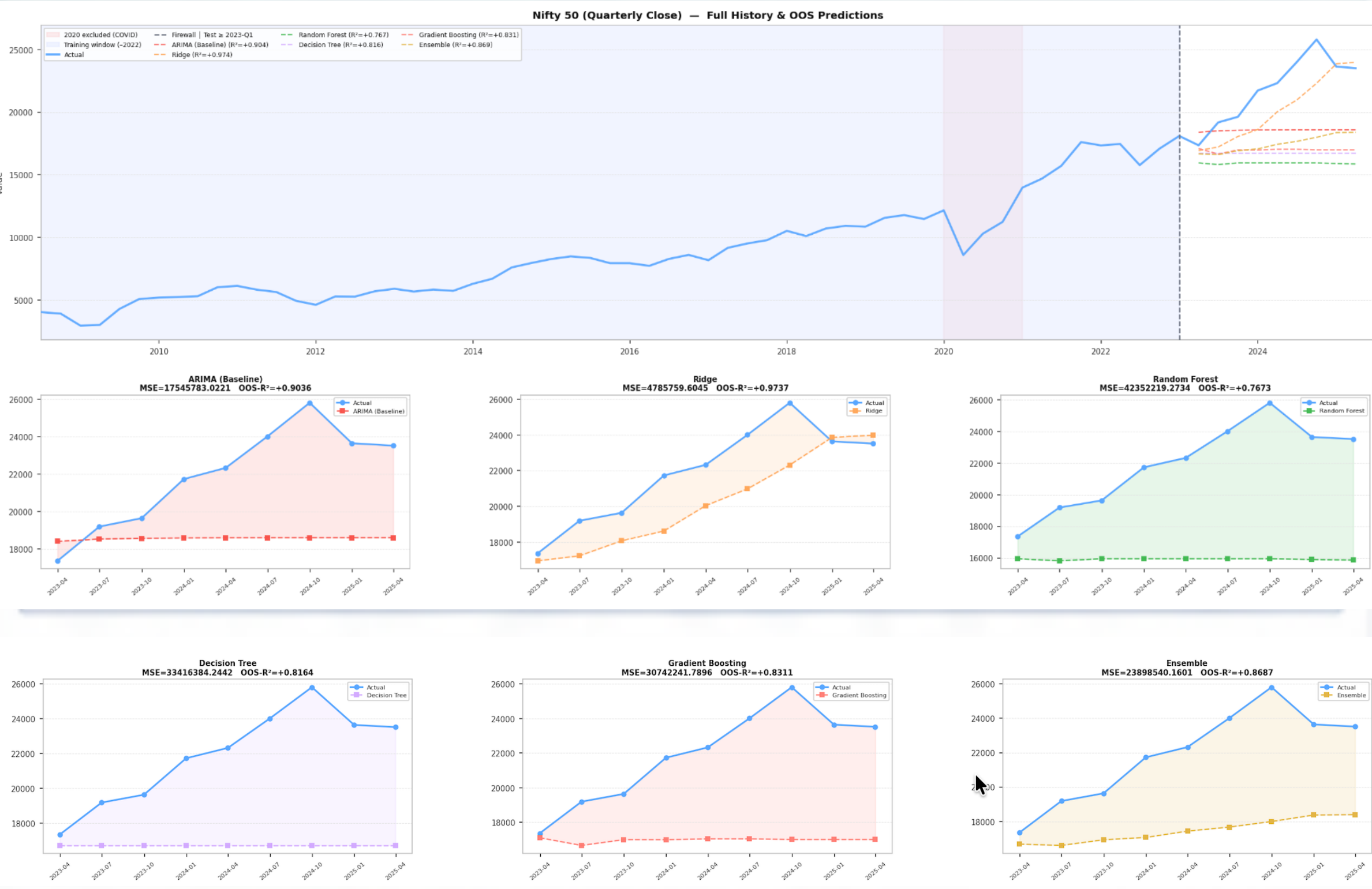
REGULARIZATION PHILOSOPHY



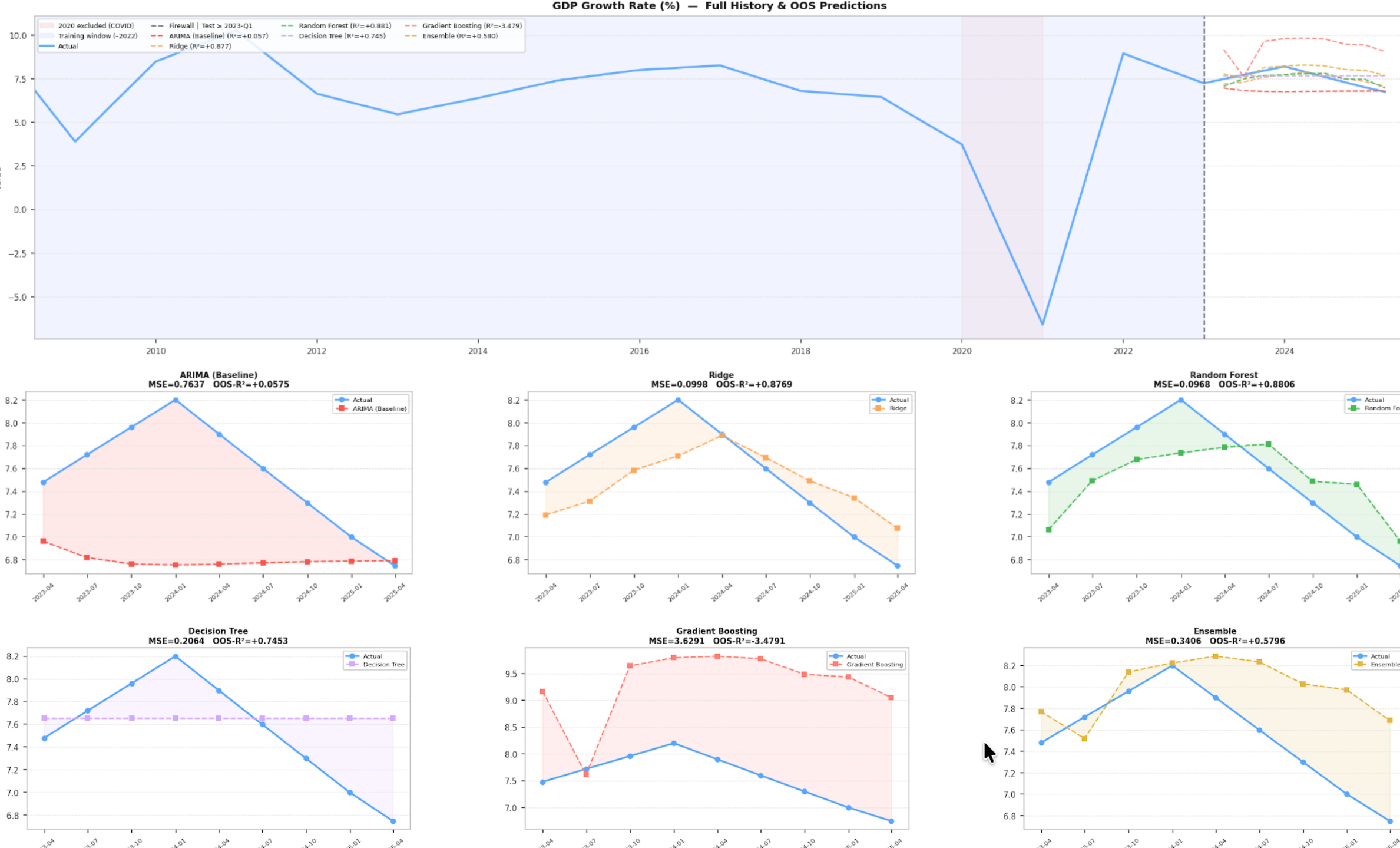
04 CORE RESULTS: MACROECONOMIC FORECASTING (QUARTERLY, 2008–2025)

GDP and CPI were linearly interpolated from annual to quarterly frequency, increasing the effective sample size to ~55 training observations.

4.1 MACRO MODELING PIPELINE (IS–LM FRAMEWORK)



4.2 GDP GROWTH RATE FORECASTS



4.3 CPI INFLATION RATE FORECASTS



KEY FINDINGS

- ★ **Ridge Regression** completely dominated Nifty 50 forecasting **OOS–R² +0.9737**
- ★ **Ridge Regression** completely dominated CPI Inflation forecasting **OOS–R² +0.9933**
- 🌲 **Random Forest** produced the best GDP Growth forecasts **OOS–R² +0.8806**
- ⚠️ **Gradient Boosting** failed catastrophically on GDP Growth forecasts **OOS–R² –3.4791**

💡 This catastrophic boosting failure proves the absolute necessity of extreme regularization (Ridge) and shallow trees to prevent overfitting in small-sample macroeconomic forecasting.

4.4 MACRO PERFORMANCE SUMMARY (OUT-OF-SAMPLE)

Target (Quarterly)	Best Model	MSE (↓)	OOS–R ² (↑)
GDP Growth Rate (%)	Random Forest	0.0986	+0.8806
CPI Inflation Rate (%)	Ridge	0.0255	+0.9933
Nifty 50 (Quarterly Close)	Ridge	478,575.6	+0.9737

Note: Lower MSE and higher R² indicate better out-of-sample performance.

05 SECONDARY RESULTS: 5-STOCK MONTHLY FORECASTING (2022–2026)

5.1 DATA CONTEXT

- 📊 Stocks: HDFCBANK, TRENT, TCS, LT, TITAN
- 📅 Training: Jan 2022 – Dec 2025 (48 months)
- 🕒 Test: Jan 2026 – May 2026 (5 months)
- 📈 Metrics: MSE (↓), OOS–R² (↑)

5.2 KEY FINDINGS

All models produced negative OOS–R² values. This is an honest result consistent with the Efficient Market Hypothesis (EMH). The short 5-month test window captured a sharp directional repricing phase in Indian large-cap equities, where past prices contain very limited predictive information about future returns.

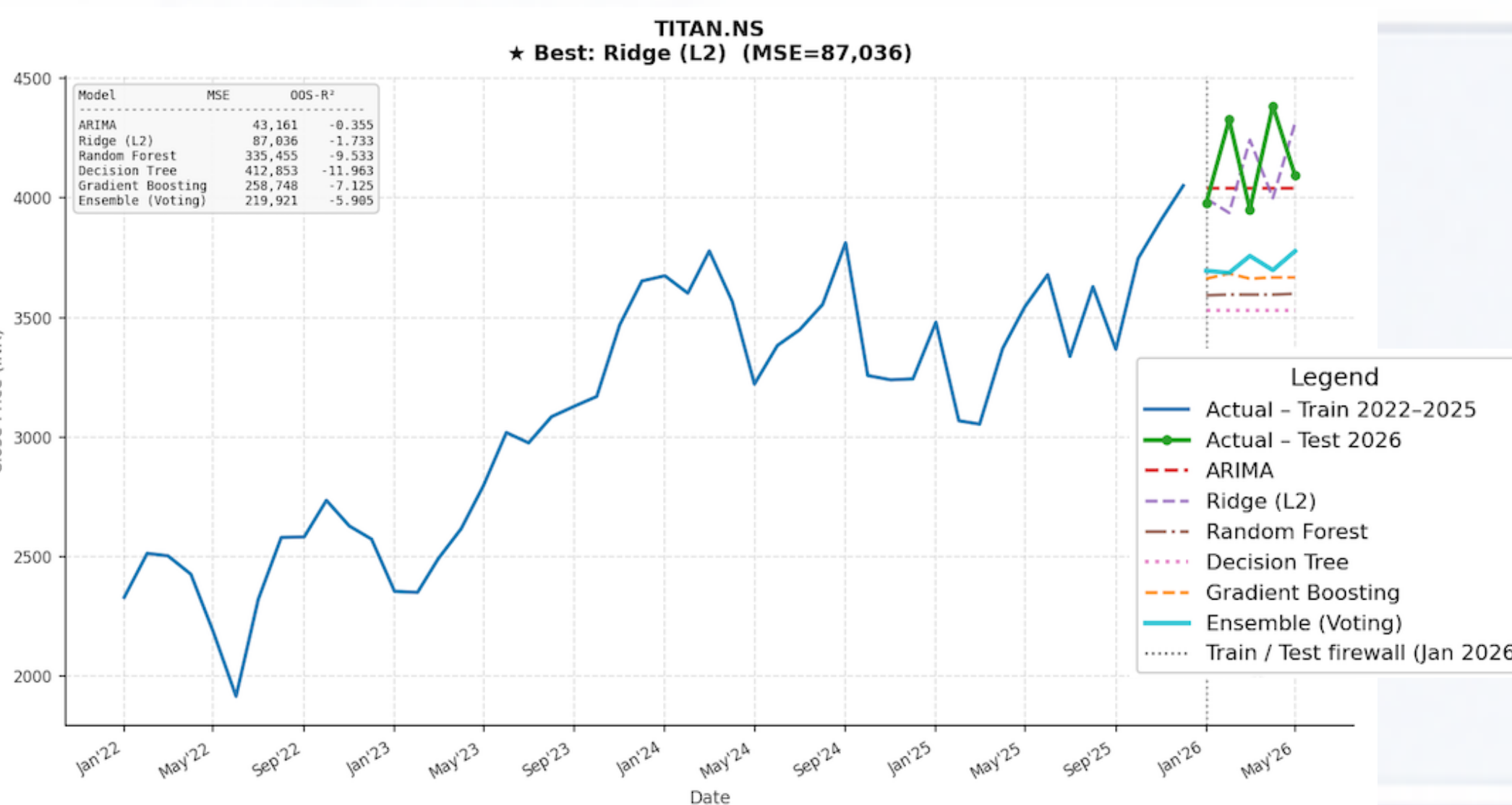


Ridge Regression emerged as the most stable ML model, winning on TRENT (OOS–R² = –0.189) and TCS (OOS–R² = –0.347) by avoiding the overfitting trap of non-linear trees.



ARIMA achieved the strongest TITAN performance (MSE = 43,161) because its (0,1,1) structure safely hedged against directional drift.

5.2 COMBINED STOCK FORECASTS (ACTUAL VS. 2026 OOS)



5.4 STOCK PERFORMANCE SUMMARY (OUT-OF-SAMPLE)

Stock	Model	MSE (↓)	OOS–R ²
HDFCBANK.NS	ARIMA	38.159	–6.068
	Ridge (L2)	7.896	–6.261
	Random Forest	16.555	–0.221
	Decision Tree	7.216	–0.121
	Gradient Boosting	7.116	–0.051
	Ensemble (Voting)	7.414	–0.183
TRENT.NS	ARIMA	738,898.97	–7.184
	Ridge (L2)	158.796	–0.190
	Random Forest	2,507.128	–25.149
	Decision Tree	2,501.228	–25.349
	Gradient Boosting	432.775	–1.140
	Ensemble (Voting)	395.975	–0.262
TCS.NS	ARIMA	414.581	–4.267
	Ridge (L2)	129.587	–0.347
	Random Forest	342.960	–6.939
	Decision Tree	248.567	–2.647
	Gradient Boosting	234.361	–1.915
	Ensemble (Voting)	263.801	–2.353
LT.NS	ARIMA	265.365	–3.316
	Ridge (L2)	166.054	–0.216
	Random Forest	162.961	–0.146
	Decision Tree	162.890	–0.160
	Gradient Boosting	160.116	–0.300
	Ensemble (Voting)	84.118	–5.316

06 CONCLUSION & ECONOMETRIC INTERPRETATION



1. Model Complexity Is Not Always Rewarded

In low-signal, small-sample economic datasets, extreme regularization (Ridge) & statistically outperforms highly flexible “black box” models.



2. Cross-Variable IS–LM Signals Matter.

ARIMA failed on all inflation forecasting (OOS–R² = –0.0647), while Ridge consistently succeeded (OOS–R² = +0.9933) because Ridge exploited cross-variable spillovers between GDP, Inflation, and Nifty dynamics that univariate ARIMA fundamentally ignores.



3. Regularization & Diversification Drive Robustness.

Shallow trees, large Ridge penalties, and firewall validation collectively prevented catastrophic overfitting and enabled out-of-sample generalization in noisy macro environments.



Project Period: Jan 2025 – May 2026